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Enhanced Bivariate FGM Copulas for Accurate Dependency Analysis

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Abstract:

A new class of bivariate FGM copulas is proposed, extending existing models. Its properties and association measures are derived, and applications to medical data show improved correlation and tail dependence. A simulation method is developed and validated to reproduce the original dependency structure.

Keywords: FGM Copula, Measures of Dependence.

Mathematics Subject Classification (2010): 62H05, 62H20.

1 Introduction

The study of copulas and their application has developed significantly in recent decades as a tool to describe the dependence of random variables (see, e.g., surveys by [Joe \(1997\)](#); [Nelsen \(2006\)](#)). Copulas play a crucial role in mathematical modeling and simulation, making the construction of diverse copula families highly meaningful. One of the most popular parametric families, studied by [Farlie \(1960\)](#), [Gumbel \(1960\)](#), and [Morgenstern \(1956\)](#), is the Farlie-Gumbel-Morgenstern (FGM) copula, defined by

$$C^{\text{FGM}}(s, t) = st [1 + \theta(1-s)(1-t)], \quad \theta \in [-1, 1], \quad \forall (s, t) \in [0, 1]^2, \quad (1.1)$$

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where θ is the association parameter. The FGM copula is PQD for $\theta \in (0, 1]$. However, this copula is somewhat limited. For the dependence parameter $\theta \in [-1, 1]$, the Spearman's rho and Kendall's tau are $\rho_S = \theta/3 \in [-1/3, 1/3]$ and $\tau_k = 2\theta/9 \in [-2/9, 2/9]$, respectively.

Due to the limited correlation range of the FGM copula, several generalizations have been proposed to improve its applicability. Symmetric semi-parametric and polynomial-type extensions were introduced by [Rodriguez-Lallena and Úbeda-Flores \(2004\)](#), [Huang and Kotz \(1999\)](#), and others, extending both correlation bounds and dependence properties. Further studies explored positive quadrant dependence, bivariate uniform distributions, and Spearman's correlation. More recent work by [Bekrizadeh et al. \(2012, 2015\)](#) and [Pathak and Vellaisamy \(2016\)](#) proposed generalized FGM copulas with wider correlation ranges, improving upon earlier FGM families.

This paper introduces a new generalized class of FGM copulas, extending recent models including [Bekrizadeh et al. \(2012\)](#), with a more efficient correlation range. Key contributions include presenting its main properties, deriving general formulas for association measures, and offering a flexible framework for modeling a wider range of dependencies. The new class enhances potential applications across various scientific fields.

2 Definition

A non-decreasing and right-continuous bivariate function $C : [0, 1]^2 \rightarrow [0, 1]$ is called a copula if it satisfies the following conditions:

$$\text{P1. } C(s, 0) = C(0, t) = 0 \text{ and } C(s, 1) = s, C(1, t) = t, \forall s, t \in [0, 1],$$

$$\text{P2. For all } s_1, s_2, t_1, t_2 \in [0, 1] \text{ with } s_1 \leq s_2 \text{ and } t_1 \leq t_2,$$

$$C(s_2, t_2) - C(s_2, t_1) - C(s_1, t_2) + C(s_1, t_1) \geq 0. \quad (2.1)$$

For twice differentiable copulas, the 2-increasing property can be replaced by the condition

$$c(s, t) = \frac{\partial^2 C(s, t)}{\partial s \partial t} \geq 0, \quad (2.2)$$

where $c(s, t)$ is the copula density.

Definition 2.1. Suppose that the strictly continuous function $\psi : [0, 1]^2 \rightarrow [0, 1]$ is differentiable on $(0, 1)$.

A function $C_{\psi, \theta}^p$ on $[0, 1]^2$ is defined as

$$C_{\psi, \theta}^p(s, t) = st [1 + \theta \psi(s, t)]^p, \quad p \in [1, \infty), \quad (s, t) \in [0, 1]^2, \quad (2.3)$$

for $\theta \in \Theta \subseteq [-1, 1]$.

Theorem 2.2. *Let ψ be differentiable, non-negative and monotonically decreasing, or non-positive and monotonically increasing on $[0, 1]$. If*

A1. $\psi(x, 0) = \psi(0, x) = 0, \forall x \in [0, 1],$

A2. i. *If $\psi'_x(x)$ and $\psi'_s(s, t)$ have the same sign for every $x \in [0, 1]$ and $(s, t) \in [0, 1]^2$, then $C_{\psi, \theta}^p$ is a copula for $\theta \in [0, 1]$; i.e., $\Theta = [0, 1]$.*

ii. *If $\psi'_x(x)$ and $\psi'_s(s, t)$ do not have the same sign for every $x \in [0, 1]$ and $(s, t) \in [0, 1]^2$, then $C_{\psi, \theta}^p$ is a copula for every $\theta \in [-1, 0]$; i.e., $\Theta = [-1, 0]$.*

Proof. The proof involves two steps. □

Corollary 2.3. *Let $\psi : [0, 1]^2 \rightarrow [0, 1]$ be differentiable and monotonic on $[0, 1]$, fulfilling the conditions*

B1. $\psi(x, 0) = \psi(0, x) = 0, \forall x \in [0, 1],$

B2. $|s\psi'_s(s, t)| \leq 1, |t\psi'_t(s, t)| \leq 1, \text{ and } |st\psi'_{st}(s, t)| \leq 1, \forall (s, t) \in [0, 1]^2.$

Then, the function $C_{\psi, \theta}^p$ is a copula for every $\theta \in [-1, 1]$; i.e., $\Theta = [-1, 1]$.

As an example, the function $\psi(s, t) = (1 - s)^\alpha(1 - t)^\alpha$, for every $\alpha \geq 0$ and the admissible range of parameter $\theta \in \left[-\frac{1}{p}, \frac{1}{p-1}\right]$, yields the family $C_{\psi, \theta}^p$ discussed by Bekrizadeh et al. (2012).

3 Key Properties

Proposition 3.1. *The family $C_{\psi, \theta}^p$ includes some known families of FGM copulas introduced by researchers in recent years, which are as follows:*

- i. *If $\psi(s, t) = (1 - s)(1 - t)$, $(s, t) \in [0, 1]^2$, and $p = 1$, the family reduces to the symmetric FGM copula discussed by Farlie (1960); Gumbel (1960); Morgenstern (1956).*
- ii. *If $\psi(s, t) = (1 - s)^\alpha(1 - t)^\alpha$, $(s, t) \in [0, 1]^2$, $\alpha \geq 0$, and $p = 1$, it reduces to the symmetric extended FGM copula introduced by Huang and Kotz (1999).*
- iii. *If $\psi(s, t) = (1 - s)^q(1 - t)^q$, $(s, t) \in [0, 1]^2$, $q \geq 1$, and $p = 1$, it reduces to the symmetric extended FGM copula introduced by Lai and Xie (2000).*
- iv. *If $\psi(s, t) = (1 - s)^\lambda(1 - t)^\gamma$, $(s, t) \in [0, 1]^2$, $\gamma \geq 0, \lambda \geq 1$, and $p = 1$, it reduces to the symmetric extended FGM copula introduced by Bairamov and Kotz (2002).*

- v. If $\psi(s, t) = \phi(s)\phi(t)$, where ϕ is defined on $[0, 1]$ such that $\phi(0) = \phi(1) = 0$, and $p = 1$, the family reduces to the class introduced by [Rodríguez-Lallena and Úbeda-Flores \(2004\)](#).
- vi. If $\psi(s, t) = (1 - s)^\alpha(1 - t)^\alpha$, $(s, t) \in [0, 1]^2$, $\alpha \geq 0$, and $p = n$, where $n = 0, 1, 2, \dots$, it reduces to the asymmetric extended FGM copula introduced by [Bekrizadeh et al. \(2012\)](#).
- vii. If $\psi(s, t) = f(s)^n g(t)^n$, $(s, t) \in [0, 1]^2$, with $f(1) = g(1) = 0$, and $p = n$, where $n = 1, 2, 3, \dots$, it reduces to the asymmetric extended FGM copula introduced by [Pathak and Vellaisamy \(2016\)](#).

Proposition 3.2. $\lim_{\theta \rightarrow 0} C_{\psi, \theta}^p(s, t) = st = \Pi(s, t)$, where $\Pi(s, t)$ is the independent copula.

4 Measures of Dependence

Measures of dependence are common instruments to summarize a complicated dependence structure in the bivariate case. For a historical review, see [Joe \(1997\)](#); [Nelsen \(2006\)](#). In this section, we compute the measures of dependence for the family $C_{\psi, \theta}^p$. Based on the binomial expansion, the family $C_{\psi, \theta}^p$ in (2.3), for every $p \in [1, \infty)$ may also be written as

$$C_{\psi, \theta}^p(s, t) = st \sum_{k=0}^p \binom{p}{k} \theta^k \psi^k(s, t) = st + st \sum_{k=1}^p \binom{p}{k} \theta^k \psi^k(s, t). \quad (4.1)$$

4.1 Spearman's rho

Let X and Y be continuous random variables whose copula is C . Then the population version of Spearman's rho is given by

$$\rho_S = 12 \int_0^1 \int_0^1 (C(s, t) - st) ds dt. \quad (4.2)$$

Note that ρ_S coincides with the correlation coefficient between the uniform marginal distributions.

Proposition 4.1. Let (X, Y) be a pair of random variables with distribution belonging to the family $C_{\psi, \theta}^p$. The Spearman's rho for the family is given by

$$\rho_S = 12 \sum_{k=1}^p \binom{p}{k} \theta^k d(\psi, k), \quad (4.3)$$

where $d(\psi, k) = \int_0^1 \int_0^1 \psi^k(s, t) ds dt$.

Proof. By using (4.1), the Spearman's rho can be expanded as

$$\begin{aligned} \rho_S &= 12 \int_0^1 \int_0^1 \left[st + st \sum_{k=1}^p \binom{p}{k} \theta^k \psi^k(s, t) - st \right] ds dt \\ &= 12 \sum_{k=1}^p \binom{p}{k} \theta^k \int_0^1 \int_0^1 st \psi^k(s, t) ds dt. \end{aligned}$$

Following the definition $d(\psi, k) = \int_0^1 \int_0^1 \psi^k(s, t) ds dt$, we obtain (4.3). $\square$

4.2 Kendall's tau

In terms of copula, Kendall's tau τ_k is defined as Nelsen (2006)

$$\tau_k = 4 \int_0^1 \int_0^1 C(s, t) c(s, t) ds dt - 1. \quad (4.4)$$

Proposition 4.2. *Let (X, Y) be a pair of random variables with the family $C_{\psi, \theta}^p$ and the density $c_{\psi, \theta}^p$; then Kendall's tau can be expanded as*

$$\tau_k = 4 \sum_{k=0}^{p-1} \binom{p}{k} \theta^{k+1} \left[(k+1) d(\psi, k+1) + \theta \sum_{j=0}^{p-k-1} \binom{p-k-1}{j} \theta^j d(\psi, k+1+j) - d(\psi, k) d(\psi, k+1) \right] - 1. \quad (4.5)$$

Proof. The proof of this proposition is deferred to Appendix B. $\square$

Example 4.3. *In the family $C_{\psi, \theta}^p$, let $\psi(s, t) = (1-s)(1-t)$. Then the family leads to a new symmetric generalized FGM copula with $\theta \in \left[-\frac{1}{p}, \frac{1}{p-1}\right]$. Since $\int_0^1 \int_0^1 (1-s)^k (1-t)^k ds dt = \frac{1}{(k+1)^2}$, we have by using (4.3) that*

$$\rho_S = 12 \sum_{k=1}^p \binom{p}{k} \frac{\theta^k}{(k+1)^2}, \quad (4.6)$$

where the upper bound of ρ_S can be increased up to approximately 0.3805 as $p \rightarrow \infty$, while the lower bound -0.3333 remains unchanged.

Example 4.4. *In the family $C_{\psi, \theta}^p$, let $\psi(s, t) = (1-s)^\alpha (1-t)^\alpha$, $\alpha \geq 0$. Then the family leads to a new symmetric generalized Huang-Kotz family with $\theta \in \left[-\frac{1}{\max\{1, p\}}, \frac{1}{p-1}\right]$. Since $\int_0^1 \int_0^1 (1-s)^{\alpha k} (1-t)^{\alpha k} ds dt = \frac{1}{(\alpha k+1)^2}$, we have*

$$\rho_S = 12 \sum_{k=1}^p \binom{p}{k} \frac{\theta^k}{(\alpha k+1)^2}. \quad (4.7)$$

Taking $\alpha \cong 1.85$ and $p > 500$ in (4.7), we have $\max \rho_S \cong 0.43$. Similarly, taking $\alpha \cong 0.1$ and $p > 500$, we obtain $\min \rho_S \cong -0.50$.

4.3 Tail Dependence

The concept of tail dependence relates to the amount of dependence in the upper-right and lower-left quadrant tails of a bivariate distribution. It is relevant for studying dependence between extreme values. For a

bivariate copula C , if

$$\lambda_U = \lim_{s \rightarrow 1^-} \frac{1 - 2s + C(s, s)}{1 - s} \quad (4.8)$$

exists, then C has upper tail dependence if $\lambda_U \in (0, 1]$, and upper tail independence if $\lambda_U = 0$. The lower tail dependence is defined similarly:

$$\lambda_L = \lim_{s \rightarrow 0^+} \frac{C(s, s)}{s}. \quad (4.9)$$

Proposition 4.5. *Let (X, Y) be a pair of random variables with distribution belonging to the family $C_{\psi, \theta}^p$; then for the copula $C_{\psi, \theta}^p$, we have*

$$\lambda_U = 1 - 2p\theta\psi(1, 1), \quad \lambda_L = 0. \quad (4.10)$$

Proof. Clearly, the upper tail dependence coefficient can be simplified using L'Hpital's rule and the derivative structure of $C_{\psi, \theta}^p(s, s)$. The lower tail dependence vanishes as $s \rightarrow 0$ due to the boundary conditions $\psi(s, 0) = \psi(0, s) = 0$. $\square$

Corollary 4.6. *Let (X, Y) be a pair of random variables with the family $C_{\psi, \theta}^p$ and $\psi(s, 1) = \psi(1, s) = 0$, for every $s \in [0, 1]$. Thus, $0 \leq \lambda_U \leq 1$ and this bound is reached within the sub-families, while for the standard FGM copula $\lambda_U = 0$.*

Example 4.7. *Choosing $\psi(s, t)$ as the cumulative distribution function of the uniform distribution on $[0, \theta]$, $\theta \leq 1$, introduced by [Huang and Kotz \(1999\)](#), gives the new family of copulas*

$$C_{\psi, \theta}^p(s, t) = st [1 + \theta \min(s, t)]^p, \quad p \in [1, \infty), \quad 0 \leq \theta \leq \frac{1}{p}.$$

By using (4.10), in this new family, we have $\lambda_U = \theta^p \in [0, 1]$.

Example 4.8. *Consider the function $\psi(s, t) = f(\min(s, t), \max(s, t))$, $(s, t) \in [0, 1]^2$, where $f(1) = 0$; this leads to a new copula. By using the tail dependence definition, in this new family, we have $\lambda_U = -\theta p f'(1)$.*

Discussion and Conclusion

This paper successfully extended the classical FGM copula family by introducing a flexible polynomial-type generator $\psi(s, t)$ and a power parameter p . The proposed framework significantly widens the admissible ranges for Spearman's rho and Kendall's tau, while enabling non-zero upper tail dependence. The theoretical properties, dependence measures, and limiting cases demonstrate the model's robustness and potential for practical applications in reliability, finance, and medical data analysis.

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